Introduction to Data Analytics on Google Cloud

Course Notes — Understanding the Data Analytics Lifecycle

Ingest | Process | Store | Analyze | Activate

Compiled from personal study notes • Google Skills: "Introduction to Data Analytics on Google Cloud"

Overview: The Data Analytics Lifecycle

Every data analytics workflow on Google Cloud follows the same basic lifecycle, no matter how big or small the project is. Three things form the **foundation** for everything else:

Storing | Managing | Organizing

On top of this foundation, data moves through five stages in a continuous loop:

INGEST → PROCESS → STORE → ANALYZE → ACTIVATE

In simple words: First you bring data in (Ingest), clean and transform it (Process), keep it somewhere safe (Store), find insights from it (Analyze), and finally put those insights to use, such as reports, dashboards, or ML models (Activate). Activate often feeds new data back into Ingest, so the cycle repeats.

Think of it like running a restaurant kitchen: ingredients arrive at the back door (Ingest), the chef washes and chops them (Process), they get stored in the fridge or pantry (Store), the chef decides what dish to cook by checking what is fresh (Analyze), and finally the dish is served to a customer (Activate). Tomorrow, new ingredients arrive again, and the cycle repeats.

Quick summary of tools used at each stage

Stage	Google Cloud Products	Purpose
Ingest	Pub/Sub, Dataflow, Dataproc, Cloud Data Fusion	Bring in real-time and batch data
Process	Dataflow, Dataproc, Cloud Data Fusion	Clean, transform, and prepare data
Store	Cloud Storage, Cloud SQL, Spanner, BigQuery, Firestore, Bigtable, AlloyDB	Keep data organized and scalable
Analyze	BigQuery, Looker, Looker Studio	Query, analyze, and visualize data
Activate	Vertex AI, AutoML, Workbench, TensorFlow	Turn insights into ML and action

Ingestion

Ingestion is the very first step: getting raw data from wherever it lives into Google Cloud, in real time or in batches.

What does "real-time" versus "batch" mean? Real-time (streaming) is like a live cricket score that updates the moment something happens. Batch is like a weekly bank statement: data is collected and processed all at once, on a schedule, for example once a day or once a week.

Cloud Data Fusion

- A **fully managed, cloud-first data integration service**.
- A **code-free** ETL (Extract, Transform, Load) tool.
- Integrates data across **on-premises and cloud sources** (on-premises simply means servers and computers owned and run by a company itself, sitting in its own office or data center, as opposed to the cloud) using a visual, drag-and-drop interface, so there is no need to write pipelines by hand.

What is ETL? ETL stands for **Extract** (pull data out of a source, like a sales system), **Transform** (clean it up by fixing errors, renaming columns, and converting formats), and **Load** (put it into its new home, like a database). Cloud Data Fusion lets you do all three by dragging and connecting boxes on a screen, similar to building a flowchart, instead of writing code.

Pub/Sub & Dataflow

- Both are **streaming analytics services** used together for real-time data pipelines.
- **Pub/Sub** is used for **ingestion**: it receives and queues messages and events as they arrive.

- **Dataflow** is used for **analytics and processing**: it transforms and processes the streamed or batched data.

In simple words: Pub/Sub is like a mailbox that catches incoming data the moment it arrives. Dataflow then opens that mail and processes it.

Process

Once data is ingested, it needs to be cleaned, transformed, and made ready for analysis. Processing simply means fixing errors, removing duplicates, converting formats, or combining fields so the data is usable.

- **Dataproc** can be used for batch processing of data.
- **Dataflow** is used to stream data for real-time processing.
- **Cloud Data Fusion** is used to integrate data coming from multiple sources into one clean pipeline.

Everyday example: Imagine raw data as vegetables straight from the farm; some are dirty, oddly shaped, or mixed with stones. Processing is washing, peeling, and cutting them into uniform pieces, so they are ready to cook (analyze) later.

Tip to remember: Dataproc means batch, Dataflow means stream, and Data Fusion means integration from many sources. All three can also help at the Ingest stage; they overlap because ingestion and processing often happen together.

Store

Storage also needs to **scale with your data**: as data grows, the storage solution should grow with it. There are three broad storage options:

Databases | Data Warehouses | Data Lakes

Storage products on Google Cloud

#	Product	Type
1	Cloud Storage	Object storage
2	Cloud SQL	Relational database
3	Cloud Spanner	Relational database
4	BigQuery	Data warehouse
5	Firestore	NoSQL database
6	Cloud Bigtable	NoSQL database
7	AlloyDB for PostgreSQL	Relational database

Databases

A database is an **organized collection of data**, stored in tables and accessed electronically from a computer system. Google Cloud offers both relational and non-relational (NoSQL) databases.

Simple analogy: A database is like a well-organized Excel workbook; data sits in neat rows and columns (tables), and you can quickly look up, add, or update information.

Relational Databases (SQL)

- Cloud SQL
- Cloud Spanner
- AlloyDB for PostgreSQL

Non-Relational Databases (NoSQL)

- Bigtable
- Firestore

Less structured, with no fixed tabular format. Follows a flexible data model, Can establish links and relationships between tables. Highly consistent and reliable, and best suited for large, **structured** data.

ideal for data whose structure changes frequently.

Simple analogy: A relational (SQL) database is like a strict Excel sheet, where every row must follow the same fixed columns (Name, Age, City). A non-relational (NoSQL) database is like a set of flexible sticky notes, where each note can hold different information, which is useful when your data does not always look the same.

Data Warehouse

- An **enterprise system** used for the analysis and reporting of structured and semi-structured data from multiple sources.
- Example on Google Cloud: **BigQuery**, used to analyze data.

Data Lake

- A repository designed to **ingest, store, explore, process, and analyze** any type or volume of raw data, regardless of source.
- Can store different types of data in their **original format**, ignoring size limits, without much pre-processing or added structure.
- Often made up of several different products, depending on the nature of the data.

Simple analogy: A **data warehouse** is like a well-labelled library, where books (data) are already sorted, catalogued, and easy to search. A **data lake** is like a large storage room where boxes of books, magazines, photos, and files are dumped as-is, unsorted, to be organized later whenever needed. Both are useful: the warehouse for quick, structured answers, and the lake for keeping everything just in case.

Types of Data

Structured | Unstructured | Semi-Structured

What do these mean? Structured data fits neatly into rows and columns, like an Excel sheet of customer names and phone numbers. **Unstructured** data has no fixed format, such as photos, videos, or free-form text. **Semi-structured** data sits in between: it has some organization, like tags or labels, but is not a strict table, such as an email that has a sender and subject field but a free-text body.

Analyze

BigQuery

- **BigQuery is at the core of data analytics on Google Cloud.**
- An **elastic, flexible, secure, and reliable data warehouse** that works across clouds and scales with your data.
- Used to analyze data through **SQL commands**.
- Lets you build a **cloud-first data warehouse**.

What is SQL? SQL, or Structured Query Language, is simply a way of asking questions to your data using near-English sentences, for example, "Show me all customers from Mumbai who spent over ₹5,000 last month." You do not need to be a programmer; SQL is one of the easiest languages to pick up, and BigQuery lets you run these questions on huge amounts of data in seconds.

Looker & Looker Studio

Used to **analyze and visualize** data, turning raw query results into charts, dashboards, and reports that are easy to understand.

Simple analogy: If BigQuery is the calculator that crunches the numbers, Looker and Looker Studio are the presenters that turn those numbers into easy-to-read graphs and dashboards, similar to how a chart is easier to understand than a raw spreadsheet.

Activate

Data is also the key to unlocking the value of **Machine Learning** on Google Cloud.

What is Machine Learning in one line? Machine Learning (ML) is teaching a computer to spot patterns in past data so it can make predictions on new data. For example, showing it thousands of past emails marked "spam" or "not spam" so it learns to automatically flag new spam emails.

Vertex AI

- The **primary development product** of the ML platform on Google Cloud.
- Vertex AI includes:

AutoML | Workbench | TensorFlow

What each piece does, in plain terms:

- **AutoML** builds machine learning models automatically, with little to no coding required, which is good for beginners.
- **Workbench** is a notebook-style workspace where data scientists write and test code for their models.
- **TensorFlow** is a popular open-source library used by developers to build custom ML models from scratch.

In simple words: Once your data is clean, stored, and analyzed, Activate is where it gets put to work: training machine learning models to make predictions or automate decisions.

Google Cloud Data Sources & Storage Methods

Data sources are **connectors** that let you query data from various sources without moving it first. Google Cloud data sources fall into two categories:

Simple analogy: Think of a data source connection like a universal charging cable. It lets BigQuery "plug into" data sitting anywhere, whether in your own Google Cloud storage or even another company's system such as AWS, and read it directly without having to copy or move the data first.

Cloud Data Sources

Stored **on Google Cloud** itself.

Example: connect to a Cloud Storage bucket or a Cloud SQL database.

External Data Sources

Stored **on-premises or in another cloud provider**.

Example: connect to an Amazon S3 bucket or a Microsoft SQL Server database.

How to create a data source connection

You can create one using:

Google Cloud Console | Cloud SDK | Google Cloud API

Benefits of Google Cloud data sources

- Gives **centralized access** to your data, making it easier to find and analyze, regardless of where it is stored.
- Helps you **integrate data** from different sources.
- Integration can be helpful for building **data warehouses and data lakes**.
- Makes it easier to **analyze** and **visualize** your data.